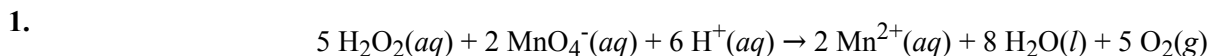


Chapter 09 Part 1: Reaction Rate



In a titration experiment, $\text{H}_2\text{O}_2(aq)$ reacts with aqueous $\text{MnO}_4^-(aq)$ as represented by the equation above. The dark purple KMnO_4 solution is added from a buret to a colorless, acidified solution of $\text{H}_2\text{O}_2(aq)$ in an Erlenmeyer flask. (Note: At the end point of the titration, the solution is a pale pink color.)

At a certain time during the titration, the rate of appearance of $\text{O}_2(g)$ was $1.0 \times 10^{-3} \text{ mol}/(\text{L}\cdot\text{s})$. What was the rate of disappearance of MnO_4^- at the same time?

(A) $6.0 \times 10^{-3} \text{ mol}/(\text{L}\cdot\text{s})$

(B) $4.0 \times 10^{-3} \text{ mol}/(\text{L}\cdot\text{s})$

(C) $6.0 \times 10^{-4} \text{ mol}/(\text{L}\cdot\text{s})$

(D) $4.0 \times 10^{-4} \text{ mol}/(\text{L}\cdot\text{s})$ ✓

2. A kinetics experiment is set up to collect the gas that is generated when a sample of chalk, consisting primarily of solid CaCO_3 , is added to a solution of ethanoic acid, CH_3COOH . The rate of reaction between CaCO_3 and CH_3COOH is determined by measuring the volume of gas generated at 25°C and 1 atm as a function of time. Which of the following experimental conditions is most likely to increase the rate of gas production?

(A) Decreasing the volume of ethanoic acid solution used in the experiment

(B) Decreasing the concentration of the ethanoic acid solution used in the experiment

(C) Decreasing the temperature at which the experiment is performed

(D) Decreasing the particle size of the CaCO_3 by grinding it into a fine powder ✓

3. A kinetics experiment is set up to collect the gas that is generated when a sample of chalk, consisting primarily of solid CaCO_3 , is added to a solution of ethanoic acid, CH_3COOH . The rate of reaction between CaCO_3 and CH_3COOH is determined by measuring the volume of gas generated at 22°C and 1 atm as a function of time. Which of the following experimental conditions is most likely to increase the rate of gas production?

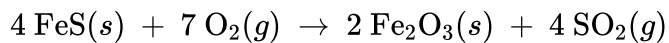
(A) Decreasing the volume of ethanoic acid solution used in the experiment

(B) Decreasing the concentration of the ethanoic acid solution used in the experiment

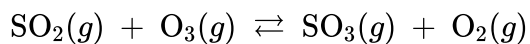
(C) Decreasing the temperature at which the experiment is performed

(D) Decreasing the particle size of the CaCO_3 by grinding it into a fine powder ✓

Chapter 09 Part 1: Reaction Rate



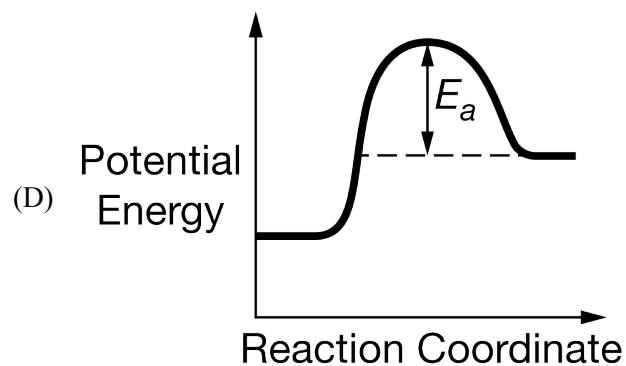
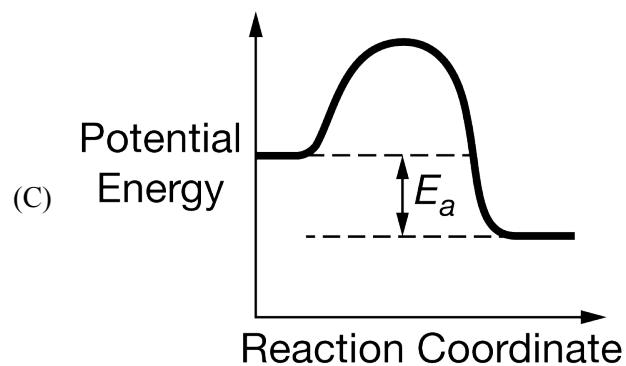
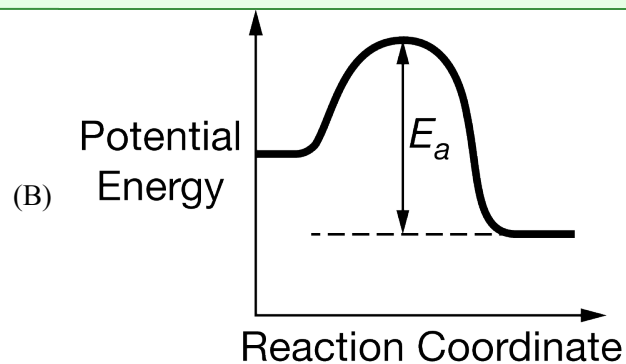
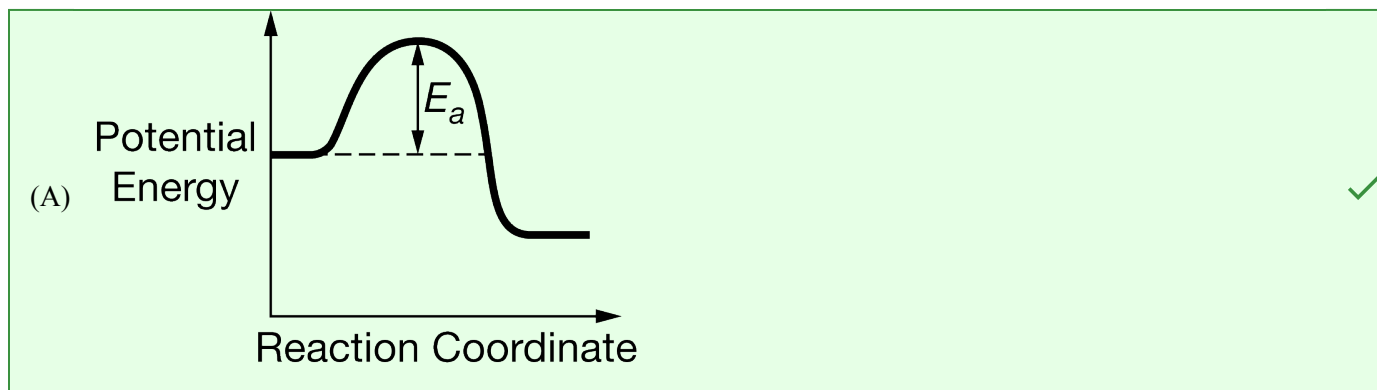
Due to the presence of $\text{FeS}(s)$ as an impurity, the combustion of some types of coal results in the formation of $\text{SO}_2(g)$, as represented by the equation above. Also, $\text{SO}_2(g)$ can react with $\text{O}_3(g)$ to form $\text{SO}_3(g)$, as represented by the equation below.



$$\Delta H_{298}^{\circ} = -242 \text{ kJ/mol}_{rxn} ; \Delta S_{298}^{\circ} = -25 \text{ J/(K} \cdot \text{mol}_{rxn})$$

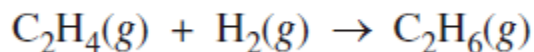
4. Which of the following diagrams shows the correctly labeled activation energy, E_a , for the reaction between $\text{SO}_2(g)$ and $\text{O}_3(g)$?

Chapter 09 Part 1: Reaction Rate



Chapter 09 Part 1: Reaction Rate

5.

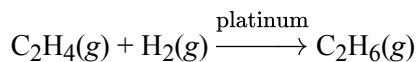


Which of the following will most likely increase the rate of the reaction represented above?

- (A) Decreasing the temperature of the reaction system
- (B) Adding a heterogeneous catalyst to the reaction system
- (C) Increasing the volume of the reaction vessel using a piston
- (D) Removing some $\text{H}_2(\text{g})$ from the reaction system



6.



$\text{C}_2\text{H}_4(\text{g})$ is reduced by $\text{H}_2(\text{g})$ in the presence of a solid platinum catalyst, as represented by the equation above. Factors that could affect the rate of the reaction include which of the following?

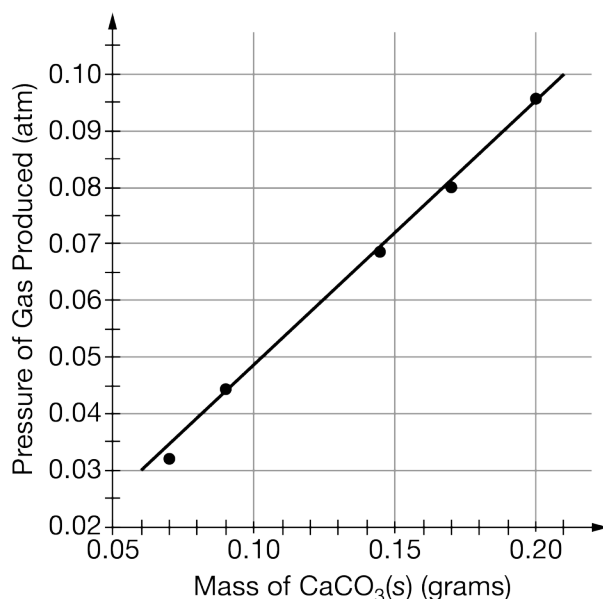
- I. Changes in the partial pressure of $\text{H}_2(\text{g})$
 - II. Changes in the particle size of the platinum catalyst
 - III. Changes in the temperature of the reaction system
- (A) III only
 - (B) I and II only
 - (C) I and III only
 - (D) II and III only

(E) I, II, and III



Chapter 09 Part 1: Reaction Rate

An experiment is performed to measure the mass percent of $\text{CaCO}_3(s)$ in eggshells. Five different samples of $\text{CaCO}_3(s)$ of known mass react with an excess of $2.0\text{ M HCl}(aq)$ in identical sealed, rigid reaction vessels. The pressure of the gas produced is measured with a pressure gauge attached to the reaction vessel. Since the reaction is exothermic, the reaction system is cooled to its original temperature before the pressure is recorded. The experimental data are used to create the calibration line below.



The experiment is repeated with an eggshell sample, and the experimental data are recorded in the table below.

Mass of eggshell sample	0.200 g
Pressure prior to reaction	0.800 atm
Pressure at completion of reaction	0.870 atm

7. Which of the following modifications will increase the rate of the reaction the most?
- (A) Using $2.0\text{ M CH}_3\text{COOH}(aq)$ instead of $2.0\text{ M HCl}(aq)$
 - (B) Cooling the $\text{HCl}(aq)$ to a lower temperature than it was in the original experiment
 - (C) Reducing the volume of the reaction vessel

(D) Using eggshells that are more finely powdered than those used in the original experiment



8. $\text{Cl}^-(aq) + \text{ClO}^-(aq) + 2\text{H}^+(aq) \rightarrow \text{Cl}_2(g) + \text{H}_2\text{O}(l)$

What effect will increasing $[\text{H}^+]$ at constant temperature have on the reaction represented above?

Chapter 09 Part 1: Reaction Rate

- (A) The activation energy of the reaction will increase.
- (B) The activation energy of the reaction will decrease.
- (C) The frequency of collisions between $\text{H}^+(aq)$ ions and $\text{ClO}^-(aq)$ ions will increase. ✓
- (D) The value of the rate constant will increase.

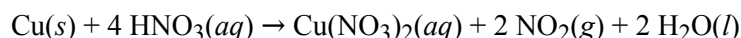
9. Two solid reactants of varying particle size are combined in a vessel. Assuming that the same mass of each reactant is used in each case, which of the following represents the conditions that will result in the fastest reaction rate?

(A) 
Temperature = 950°C

(B) 
Temperature = 500°C

(C) 
Temperature = 500°C

(D) 
Temperature = 950°C ✓

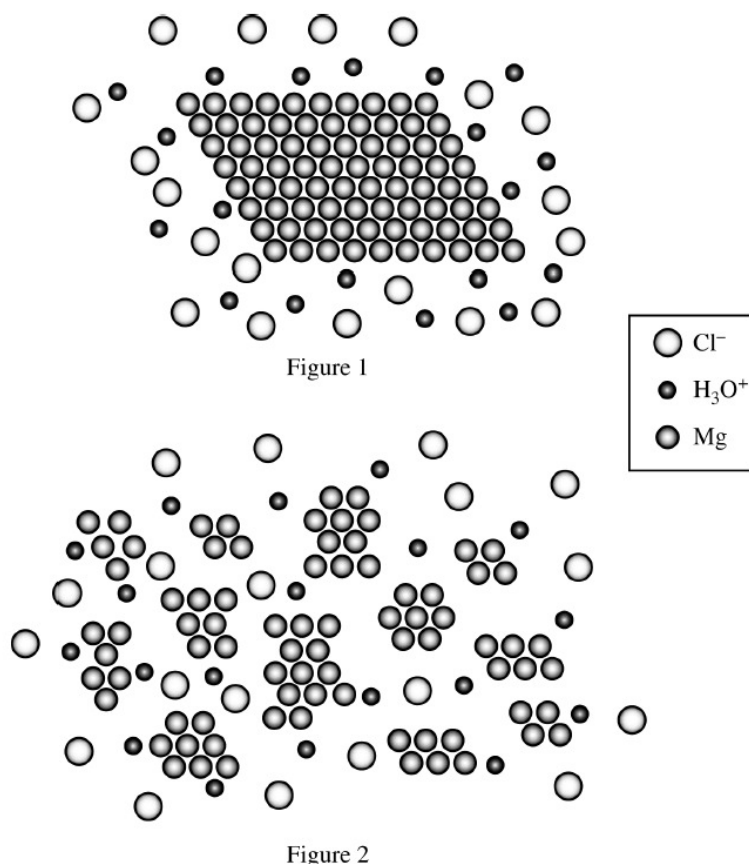


Each student in a class placed a 2.00 g sample of a mixture of Cu and Al in a beaker and placed the beaker in a fume hood. The students slowly poured 15.0 mL of 15.8 M $\text{HNO}_3(aq)$ into their beakers. The reaction between the copper in the mixture and the $\text{HNO}_3(aq)$ is represented by the equation above. The students observed that a brown gas was released from the beakers and that the solutions turned blue, indicating the formation of $\text{Cu}^{2+}(aq)$. The solutions were then diluted with distilled water to known volumes.

10. In one student's experiment the reaction proceeded at a much slower rate than it did in the other students' experiments. Which of the following could explain the slower reaction rate?
- (A) In the student's sample the metal pieces were much smaller than those in the other students' samples.
- (B) The student heated the reaction mixture as the $\text{HNO}_3(aq)$ was added.
- (C) The student used a 1.5 M solution of $\text{HNO}_3(aq)$ instead of a 15.8 M solution of $\text{HNO}_3(aq)$. ✓
- (D) The student used a 3.00 g sample of the mixture instead of the 2.00 g sample that was used by the other students.

Chapter 09 Part 1: Reaction Rate

11.



Two samples of $\text{Mg}(s)$ of equal mass were placed in equal amounts of $\text{HCl}(aq)$ contained in two separate reaction vessels. Particle representations of the mixing of $\text{Mg}(s)$ and $\text{HCl}(aq)$ in the two reaction vessels are shown in Figure 1 and Figure 2 above. Water molecules are not included in the particle representations. Which of the reactions will initially proceed faster, and why?

- (A) The reaction in Figure 1, because the atoms of Mg are more concentrated than those in Figure 2
- (B) The reaction in Figure 1, because the $\text{Mg}(s)$ in Figure 1 has a larger mass than the $\text{Mg}(s)$ in Figure 2
- (C) The reaction in Figure 2, because more Mg atoms are exposed to $\text{HCl}(aq)$ in Figure 2 than in Figure 1 ✓
- (D) The reaction in Figure 2, because the $\text{Mg}(s)$ in Figure 2 has less surface area than the $\text{Mg}(s)$ in Figure 1

A 0.50 g sample of $\text{Mg}(s)$ was placed in a solution of $\text{HCl}(aq)$, where it reacted completely.

12. In a third experiment, 0.10 g samples of $\text{Mg}(s)$ are placed in excess $\text{HCl}(aq)$ of various concentrations: 0.050 M, 0.10 M, 0.25 M, and 0.50 M. The reactions are run in successive order from 0.050 M to 0.50 M, and the time required for each reaction to go to completion is recorded. As the concentration of $\text{HCl}(aq)$ increases from 0.050 M to 0.50 M, which of the following is the expected result?

Chapter 09 Part 1: Reaction Rate

(A) The reaction time increases, and the rate of the reaction decreases.

(B) The reaction time decreases, and the rate of the reaction increases. ✓

(C) Both the reaction time and the rate of the reaction increase.

(D) Both the reaction time and the rate of the reaction decrease.

13. The gas-phase reaction $A_2(g) + B_2(g) \rightarrow 2 AB(g)$ is assumed to occur in a single step. Two experiments were done at the same temperature inside rigid containers. The initial partial pressures of A_2 and B_2 used in experiment 1 were twice the initial pressures used in experiment 2. Which statement provides the best comparison of the initial rate of formation of AB in experiments 1 and 2 ?

(A) The initial rate of formation of AB is the same in both experiments because they were done at the same temperature and the frequency and energy of the collisions between A_2 and B_2 would have been about the same.

(B) The initial rate of formation of AB is slower in experiment 1 than in with experiment 2 because at the same temperature, a higher pressure would reduce the volume available for A_2 and B_2 molecules to achieve the proper orientation for a successful collision.

(C) The initial rate of formation of AB is faster in experiment 1 than in experiment 2 because at a higher pressure the collisions between A_2 and B_2 molecules would have been more frequent, increasing the probability of a successful collision. ✓

(D) The initial rate of formation of AB is faster in experiment 1 than in experiment 2 because at a higher pressure a larger fraction of the A_2 and B_2 molecules would have the minimum energy required to overcome the activation energy barrier.

14. Which of the following best helps explain why an increase in temperature increases the rate of a chemical reaction?

(A) At higher temperatures, reactions have a lower activation energy.

(B) At higher temperatures, reactions have a higher activation energy.

(C) At higher temperatures, every collision results in the formation of product.

(D) At higher temperatures, high-energy collisions happen more frequently. ✓

15. Factors that affect the rate of a chemical reaction include which of the following?

I. Frequency of collisions of reactant particles

II. Kinetic energy of collisions of reactant particles

III. Orientation of reactant particles during collisions

(A) II only

(B) I and II only

(C) I and III only

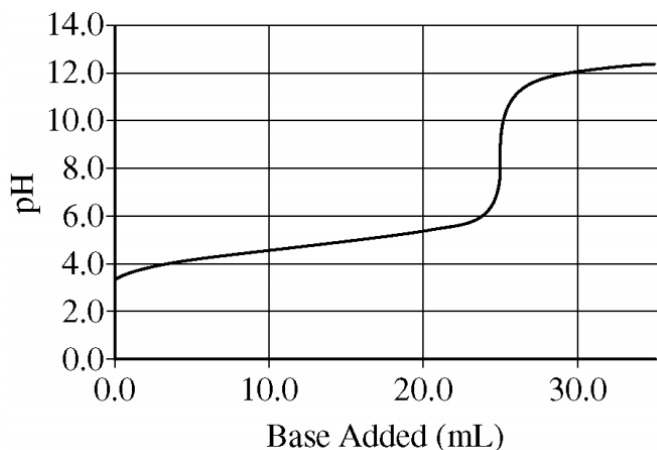
(D) II and III only

(E) I, II, and III ✓

Chapter 09 Part 1: Reaction Rate

A 0.35 g sample of $\text{Li}(s)$ is placed in an Erlenmeyer flask containing 100 mL of water at 25°C . A balloon is placed over the mouth of the flask to collect the hydrogen gas that is generated.

After all of the $\text{Li}(s)$ has reacted with $\text{H}_2\text{O}(l)$, the solution in the flask is added to a clean, dry buret and used to titrate an aqueous solution of a monoprotic acid. The pH curve for this titration is shown in the diagram below

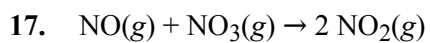


16. Which of the following changes will most likely increase the rate of reaction between $\text{Li}(s)$ and water?

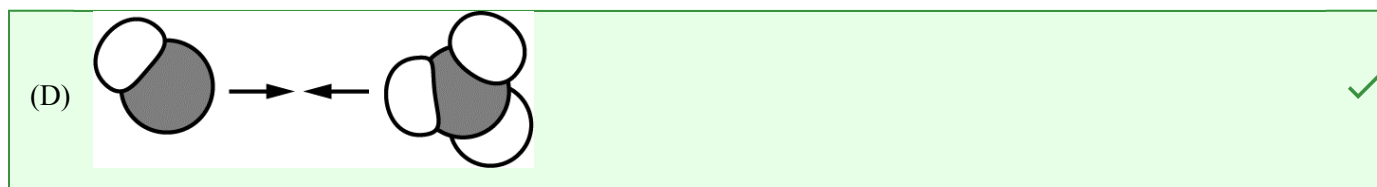
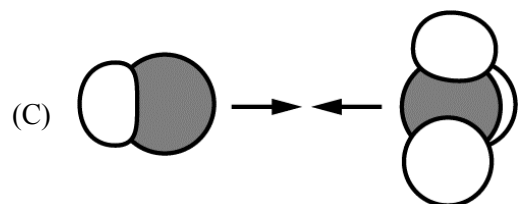
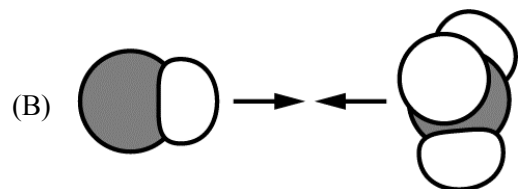
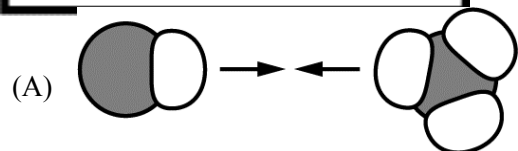
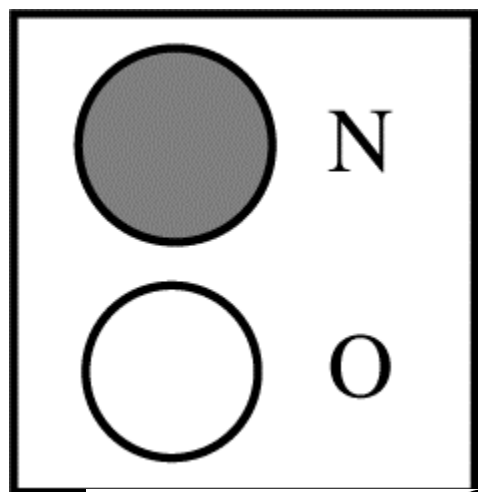
- (A) Using 125 mL of water instead of 100 mL
- (B) Using a 0.25 g sample of $\text{Li}(s)$ instead of a 0.35 g sample
- (C) Using a 0.35 g sample of $\text{Li}(s)$ cut into small pieces
- (D) Decreasing the water temperature before adding the $\text{Li}(s)$



Chapter 09 Part 1: Reaction Rate

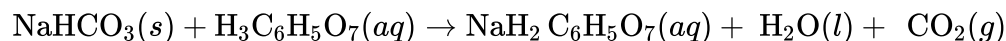


The reaction between $\text{NO}(g)$ and $\text{NO}_3(g)$ is represented by the equation above. Which of the following orientations of collision between $\text{NO}(g)$ and $\text{NO}_3(g)$ is most likely to be effective?



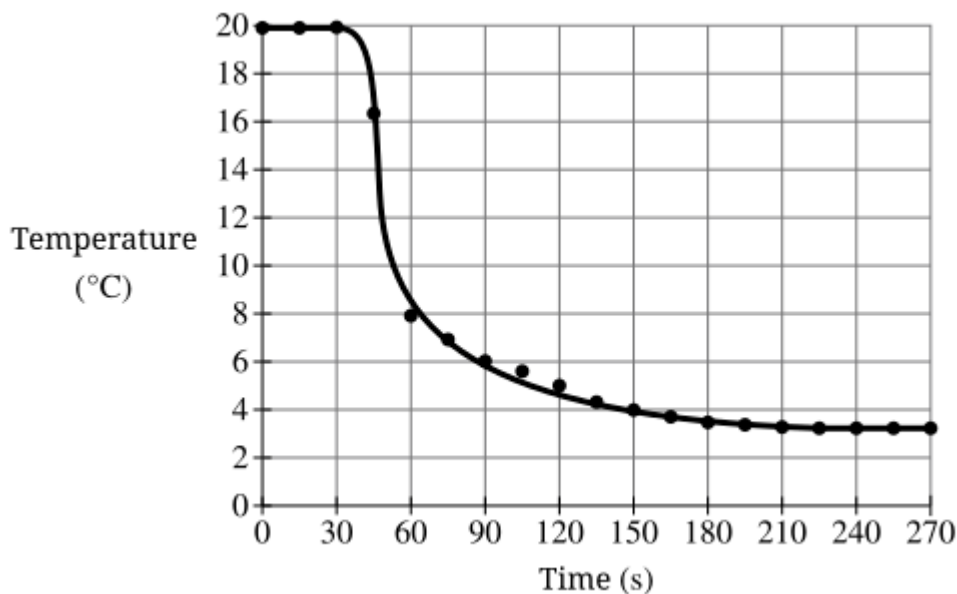
Chapter 09 Part 1: Reaction Rate

18. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



A student conducts a calorimetry experiment in which solid baking soda, NaHCO_3 , reacts with excess aqueous citric acid, $\text{H}_3\text{C}_6\text{H}_5\text{O}_7$, as represented by the equation above. The student places 36.40 g of the $\text{H}_3\text{C}_6\text{H}_5\text{O}_7$ solution into a calorimeter and measures the temperature. After 30 seconds, 8.00 g of NaHCO_3 is added. The data are recorded in the following table and graph.

Mass of $\text{H}_3\text{C}_6\text{H}_5\text{O}_7(aq)$	36.40 g
Mass of $\text{NaHCO}_3(s)$	8.00 g
Total mass of reaction mixture	44.40 g
Initial temperature	19.9 °C
Final temperature	3.2 °C



- (a) Calculate the magnitude of q , in joules, for this mixture using the data in the table. Assume that the reaction mixture in the calorimeter has a specific heat capacity of $4.22 \text{ J}/(\text{g} \cdot ^\circ\text{C})$.
- (b) Based on the collisions that occur between particles, explain why the graph shows temperature changing less rapidly as time progresses in the experiment.

Chapter 09 Part 1: Reaction Rate

(c) Make a claim about the flow of heat between the reaction system and the surroundings during this experiment. The claim should be consistent with the data in the graph.

(d) The student conducts a second experiment using 36.40 g of $\text{H}_3\text{C}_6\text{H}_5\text{O}_7(aq)$ at an initial temperature of 19.9°C . However, the student adds only 4.00 g of $\text{NaHCO}_3(s)$. Would the temperature change in the second experiment be greater than, less than, or equal to the temperature change in the first experiment? Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct calculated value.

$$\bullet \quad q = (44.40 \text{ g})(4.22 \text{ J}/(\text{g} \cdot ^\circ\text{C}))(3.2^\circ\text{C} - 19.9^\circ\text{C})$$

$q = -3130 \text{ J}$. Therefore, the magnitude of q is 3130 J.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a correct explanation.

Chapter 09 Part 1: Reaction Rate

- As the reaction progresses, a lower concentration of the reactant particles remains in the solution, resulting in fewer collisions and a reduced rate of temperature change.

OR

- As the temperature decreases, collisions between reactant particles occur less frequently and with less force. Since energy must be absorbed for the reaction to occur, having a lower temperature in the solution means fewer particles will achieve the energy requirements to produce products.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a correct claim.

- Heat transfers from the surroundings into the system.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a correct prediction.

- Less than. The temperature change will be smaller because less of the limiting reactant is present, resulting in a smaller thermal energy transfer (smaller value of q).



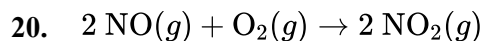
$$\text{rate} = k[\text{NO}][\text{NO}_3]$$

The reaction represented above occurs in a single step that involves the collision between a particle of NO and a particle of NO₃. A scientist correctly calculates the rate of collisions between NO and NO₃ that have sufficient energy to overcome the activation energy. The observed reaction rate is only a small fraction of the calculated collision rate. Which of the following best explains the discrepancy?

Chapter 09 Part 1: Reaction Rate

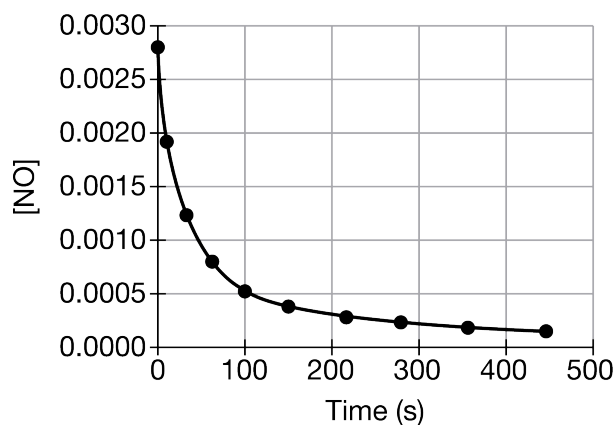
- (A) The energy of collisions between two reactant particles is frequently absorbed by collision with a third particle.
- (B) The two reactant particles must collide with a particular orientation in order to react. ✓
- (C) The activation energy for a reaction is dependent on the concentrations of the reactant particles.
- (D) The activation energy for a reaction is dependent on the temperature.

Chapter 09 Part 1: Reaction Rate

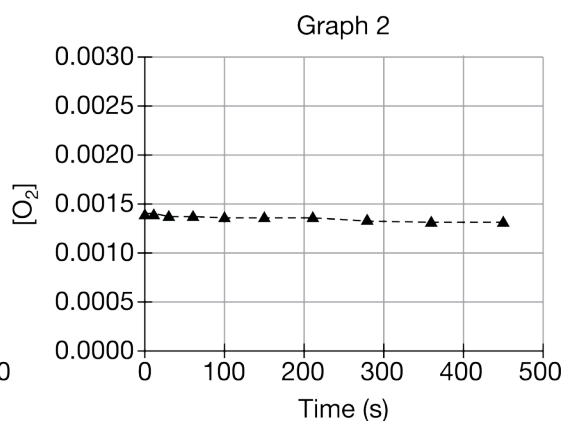
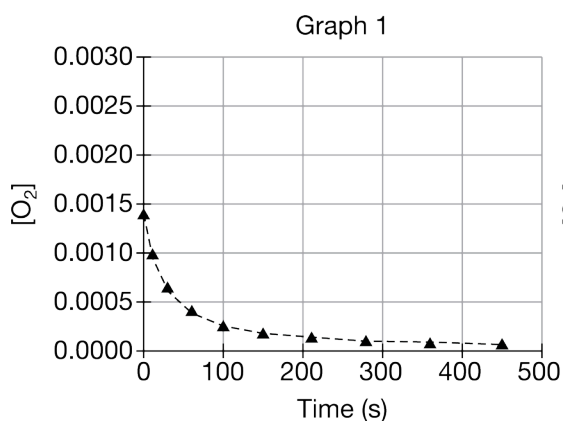


Reactant	Initial Concentration
NO	0.0028 M
O ₂	0.0014 M

The oxidation of $\text{NO}(g)$ producing $\text{NO}_2(g)$ is represented by the chemical equation shown above. The initial concentrations of NO and O_2 are given in the table above. The changes in the concentration of $\text{NO}(g)$ as a function of time are shown in the following graph.



Under the same conditions, one of the following graphs represents the changes in the concentration of $\text{O}_2(g)$ over the same period of time.

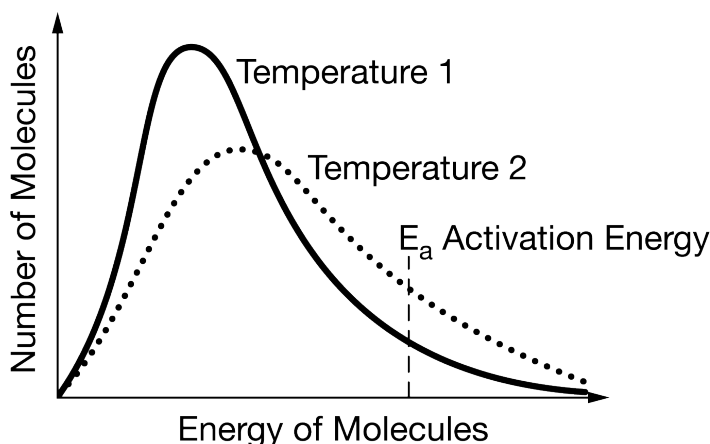


Which graph best represents the changes in concentration of $\text{O}_2(g)$, and why?

Chapter 09 Part 1: Reaction Rate

- (A) Graph 1, because the rate of O_2 consumption is half the rate at which NO is consumed; two molecules of NO react for each molecule of O_2 that reacts. ✓
- (B) Graph 1, because O_2 molecules are consumed at a slower rate at the beginning of the reaction when there are not as many molecules of NO_2 produced.
- (C) Graph 2, because there is a large excess of O_2 molecules and its concentration will not change drastically over time.
- (D) Graph 2, because a collision between three molecules to form a product has a low probability and the concentration of O_2 will remain relatively constant.

21.

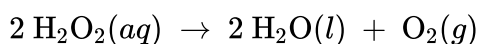


The proposed rate-determining step for a reaction is $2\text{NO}_2(g) \rightarrow \text{NO}_3(g) + \text{NO}(g)$. The graph above shows the distribution of energies for $\text{NO}_2(g)$ molecules at two temperatures. Based on the graph, which of the following statements best explains why the rates of disappearance of $\text{NO}_2(g)$ are different at temperature 2 and temperature 1?

- (A) $\text{NO}_2(g)$ is consumed at a faster rate at temperature 2 because more molecules possess energies at or above the minimum energy required for a collision to lead to a reaction compared to temperature 1. ✓
- (B) $\text{NO}_2(g)$ is consumed at a faster rate at temperature 2 because the molecules have a wider range of energies allowing for a better orientation during a collision compared to temperature 1.
- (C) Fewer $\text{NO}_2(g)$ molecules have a relatively high energy at temperature 1, which favors collisions between molecules rather than between the molecules and the container, leading to a faster rate of disappearance compared to temperature 2.
- (D) More $\text{NO}_2(g)$ molecules have a relatively low energy at temperature 1, which increases the number of effective collisions taking place and the rate of disappearance compared to temperature 2.

Chapter 09 Part 1: Reaction Rate

The following balanced equation represents the decomposition of hydrogen peroxide, H_2O_2 .

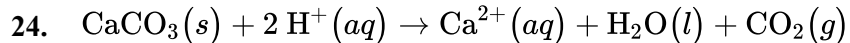


A student carries out an experiment in which a sample of $\text{H}_2\text{O}_2(aq)$ decomposes in the presence of $\text{MnO}_2(s)$. The student measures the volume of $\text{O}_2(g)$ produced by the reaction at regular intervals while the temperature is held constant. The student's data are given in the following table.

Time (s)	Volume of $\text{O}_2(g)$ Collected (mL)
0	0
60	42
120	64
180	76
240	82
300	86
360	88
420	89

22. Based on the student's data, which of the following statements correctly describes how the rate of O_2 production changes during the experiment?
- (A) The rate is constant at first and then increases over time.
- (B) The rate decreases throughout the experiment. ✓
- (C) The rate increases throughout the experiment.
- (D) The rate is constant throughout the experiment.
23. A second experiment is done under identical conditions using a higher concentration of $\text{H}_2\text{O}_2(aq)$ in the presence of the same mass of $\text{MnO}_2(s)$. The volume of gas collected during the first 60 seconds will most likely be
- (A) less than 42 mL
- (B) equal to 42 mL
- (C) greater than 42 mL ✓
- (D) impossible to predict without knowing the actual concentration of H_2O_2 used in the two experiments
-

Chapter 09 Part 1: Reaction Rate

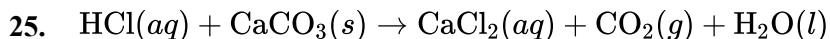


Experiment	Initial Mass of $\text{CaCO}_3(s)$ (grams)	Initial Concentration of $\text{H}^+(aq)$ (M)	Initial Temperature of Reactants ($^{\circ}\text{C}$)
1	5.00	1.0	25.0
2	5.00	2.0	25.0

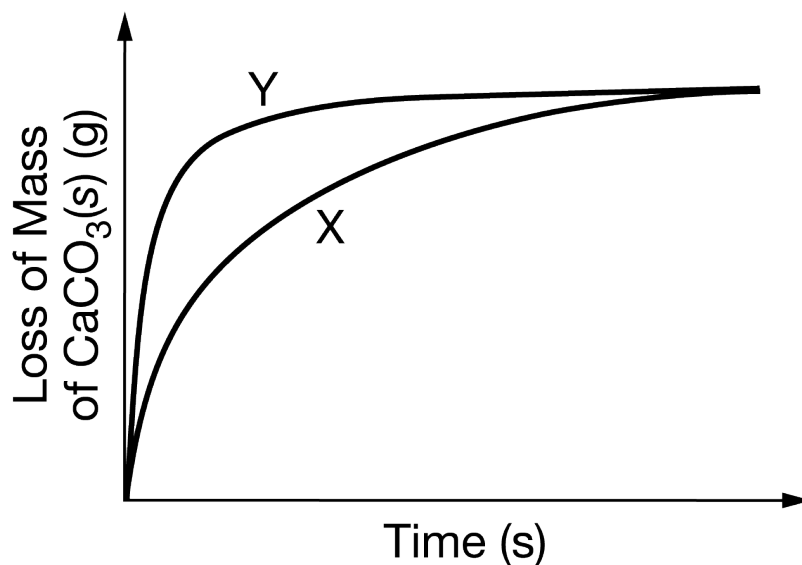
The initial rate of formation of $\text{CO}_2(g)$ from the chemical reaction represented by the equation above was studied in two separate experiments. The table above provides the experimental conditions used. If both experiments are carried out with finely powdered samples of the solid and 50.0 mL of $\text{HCl}(aq)$, which experiment, if any, will have the faster initial rate of formation of $\text{CO}_2(g)$ and why?

- (A) The rate of formation of $\text{CO}_2(g)$ will be the same because the mass of $\text{CaCO}_3(s)$ and the volume of $\text{HCl}(aq)$ used will be the same in both experiments.
- (B) The rate of formation of $\text{CO}_2(g)$ will be the same because the surface area of the solid and the average kinetic energy of the particles will be the same in both experiments.
- (C) $\text{CO}_2(g)$ will be formed at a faster rate in experiment 2 because more H^+ particles can react per unit time. ✓
- (D) $\text{CO}_2(g)$ will be formed at a faster rate in experiment 1 because the proportion of $\text{CaCO}_3(s)$ particles to H^+ particles will be greater.

Chapter 09 Part 1: Reaction Rate



The reaction between $\text{HCl}(aq)$ and $\text{CaCO}_3(s)$ is represented by the equation above. Two separate trials were carried out using $\text{CaCO}_3(s)$ samples of the same mass, but one sample was a single piece of $\text{CaCO}_3(s)$, and one sample was composed of small pieces of $\text{CaCO}_3(s)$. The loss of mass of $\text{CaCO}_3(s)$ as a function of time for both trials is shown in the graph below.



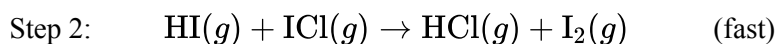
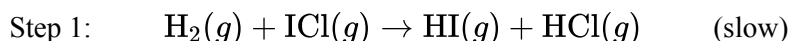
Which of the curves, X or Y, represents the reaction with small pieces of $\text{CaCO}_3(s)$, and why?

- (A) Curve X, because it shows that the reaction proceeded at a uniform rate.
- (B) Curve X, because it takes a shorter time for the reaction to go to completion due to the larger surface area of $\text{CaCO}_3(s)$.
- (C) Curve Y, because it shows that the reaction proceeded at a nonuniform rate.
- (D) Curve Y, because it takes a shorter time for the reaction to go to completion due to the larger surface area of $\text{CaCO}_3(s)$. ✓

26. When the chemical reaction $2\text{NO}(g) + \text{O}_2(g) \rightarrow 2\text{NO}_2(g)$ is carried out under certain conditions, the rate of disappearance of $\text{NO}(g)$ is $5.0 \times 10^{-5} \text{ Ms}^{-1}$. What is the rate of disappearance of $\text{O}_2(g)$ under the same conditions?

- (A) Because two molecules of NO are consumed per molecule of O_2 , the rate of disappearance of $\text{O}_2(g)$ is $2.5 \times 10^{-5} \text{ Ms}^{-1}$. ✓
- (B) Because two molecules of NO are consumed per molecule of O_2 , the rate of disappearance of $\text{O}_2(g)$ is $1.0 \times 10^{-4} \text{ Ms}^{-1}$.
- (C) Because the rate depends on $[\text{NO}]^2$ and $[\text{O}_2]$, the rate of disappearance of $\text{O}_2(g)$ is $2.5 \times 10^{-9} \text{ Ms}^{-1}$.
- (D) Because the $\text{O}_2(g)$ is under the same experimental conditions as NO , it is consumed at the same rate of $5.0 \times 10^{-5} \text{ Ms}^{-1}$.

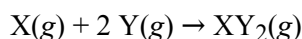
Chapter 09 Part 1: Reaction Rate



27. The reaction is carried out at constant temperature inside a rigid container. Based on this mechanism, which of the following is the most likely reason for the different rates of step 1 and step 2 ?

- (A) The only factor determining the rate of step 2 is the orientation of the HI and ICl polar molecules during a collision, but it has a negligible effect when H₂ and ICl molecules collide.
- (B) The amount of energy required for a successful collision between H₂ and ICl is greater than the amount of energy required for a successful collision between HI and ICl. ✓
- (C) The fraction of molecules with enough energy to overcome the activation energy barrier is lower for HI and ICl than for H₂ and ICl.
- (D) The frequency of collisions between H₂ and ICl is greater than the frequency of collisions between HI and ICl.
-

28.



In order to determine the order of the reaction represented above, the initial rate of formation of XY₂ is measured using different initial values of [X] and [Y]. The results of the experiment are shown in the table below.

Trial	[X]	[Y]	Initial Rate of Formation of XY ₂ (M s ⁻¹)
1	0.50	0.50	8.0×10^{-3}
2	1.00	0.50	3.2×10^{-2}
3	1.00	1.00	6.4×10^{-2}

In trial 2 which of the reactants would be consumed more rapidly, and why?

- (A) X, because it has a higher molar concentration.
- (B) X, because the reaction is second order with respect to X.
- (C) Y, because the reaction is second order with respect to Y.
- (D) Y, because the rate of disappearance will be double that of X. ✓

29. Relatively slow rates of chemical reaction are associated with which of the following?

Chapter 09 Part 1: Reaction Rate

- (A) The presence of a catalyst
- (B) High temperature
- (C) High concentration of reactants
- (D) Strong bonds in reactant molecules ✓
- (E) Low activation energy

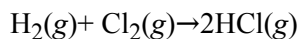
30. Which of the following best describes the role of the spark from the spark plug in an automobile engine?

- (A) The spark decreases the energy of activation for the slow step.
- (B) The spark increases the concentration of the volatile reactant.
- (C) The spark supplies some of the energy of activation for the combustion reaction. ✓
- (D) The spark provides a more favorable activated complex for the combustion reaction.
- (E) The spark provides the heat of vaporization for the volatile hydrocarbon.

31.

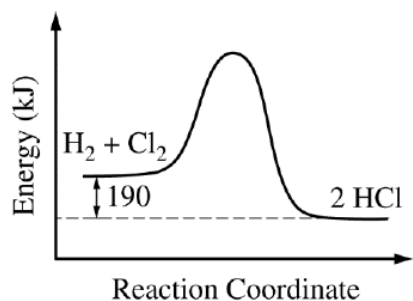
Bond	Bond Energy (kJ/mol)
H–H	430
Cl–Cl	240
H–Cl	430

Based on the bond energies shown in the table above, which of the following diagrams best represents the change in energy as the reaction represented below proceeds?

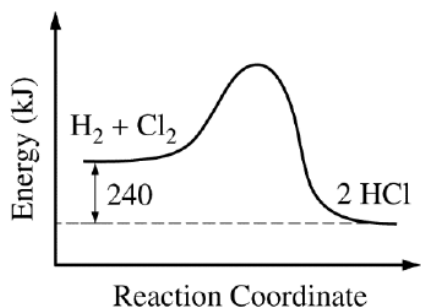


Chapter 09 Part 1: Reaction Rate

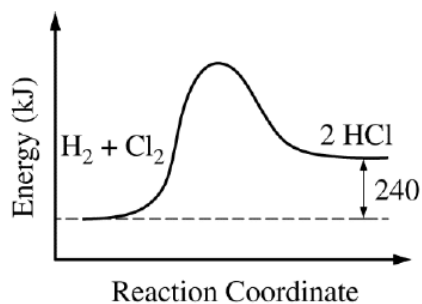
(A)



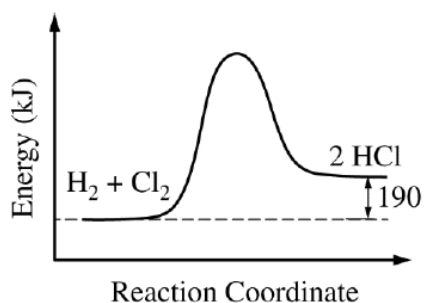
(B)



(C)



(D)



32. Which of the following statements best explains why an increase in temperature of 5-10 Celsius degrees can substantially increase the rate of a chemical reaction?

(A) The activation energy for the reaction is lowered.

(B) The number of effective collisions between reactant particles is increased.



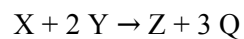
(C) The rate of the reverse reaction is increased.

(D) ΔH for the reaction is lowered.

(E) ΔG for the reaction becomes more positive.

Chapter 09 Part 1: Reaction Rate

33.



For the reaction represented above, the initial rate of decrease in $[\text{X}]$ was $2.8 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$. What was the initial rate of decrease in $[\text{Y}]$?

(A) $7.0 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$

(B) $1.4 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$

(C) $2.8 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$

(D) $5.6 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$

(E) $1.1 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$

